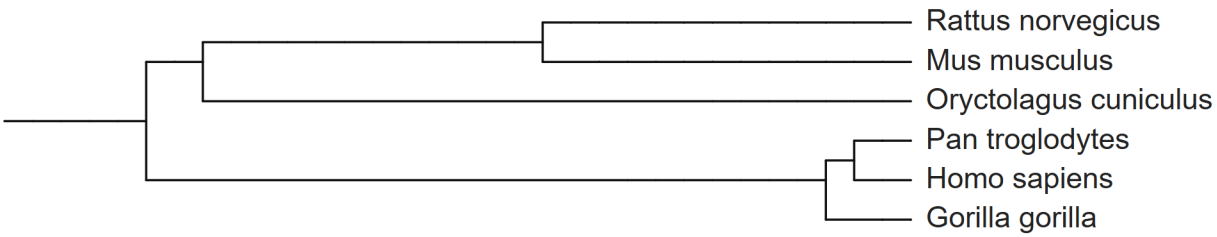
Common ancestry and clades

Trace shared ancestors and recognize nested groups in a small selection from a published tree.

Work from branching connections, not the apparent distance between names. These six mammals are a selection from a larger source; branch spacing does not measure time.

Name: _____ Date: _____



Six selected mammals from Upham et al. (2019), S3 global ML tree, Dryad version 4. Nonmetric cladogram; original source retained. Data: CC0.

Text description: Within the selected source, human and chimpanzee form a represented pair nested with gorilla. Mouse and rat form a separate represented pair; rabbit is also included. Other descendants exist in the full source and are omitted from the drawing.

1. Trace Homo sapiens and Pan troglodytes to their most recent common ancestor. Is Gorilla gorilla descended from that same most recent node?

2. Identify the represented group containing human, chimpanzee and gorilla. Explain why leaving chimpanzee out would make the proposed grouping incomplete.

3. Find a second represented pair that shares a common ancestor excluding the other four displayed taxa. What additional information would you need before calling either pair a complete clade in the full source?

Upham, Esselstyn & Jetz (2019). Paper: 10.1371/journal.pbio.3000494. Data: 10.5061/dryad.tb03d03, version 4, S3. Activity: PhyloInk education, 0BSD.